

FEDERATED GRAPH NEURAL NETWORK FOR MULTI-DOMAIN MENTAL HEALTH DIGITAL TWIN GENERATION

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ABSTRACT

A system for generating and maintaining a multi-domain digital twin of a patient's mental health state using a Graph Neural Network (GNN). The digital twin represents 12 distinct life domains as nodes in a graph, with edges representing learned probabilistic dependencies. The system employs a federated learning architecture with differential privacy noise injection, allowing the GNN to be trained across multiple disparate healthcare institutions without transmitting Protected Health Information (PHI). An Isolation Forest algorithm continuously monitors the high-dimensional graph state to detect complex, multi-domain anomalies indicative of psychiatric deterioration.

BACKGROUND OF THE INVENTION

While digital twins exist in industrial IoT and general healthcare (e.g., US20250356078A1), no existing patent specifically claims a Graph Neural Network architecture for modeling the interdependent domains of mental health. Furthermore, training such complex models typically requires centralizing highly sensitive psychiatric data, violating HIPAA and privacy norms. Existing anomaly detection fails to capture multi-domain interactions (e.g., a slight drop in sleep combined with a slight drop in social activity that together indicate severe risk).

SUMMARY OF THE INVENTION

The present invention introduces a HIPAA-compliant, federated GNN architecture specifically for mental health. By representing domains (Sleep, Mood, Anxiety, etc.) as nodes in a graph, the GNN's message-passing layers explicitly learn how a deficit in one domain propagates to others. The federated learning protocol with differential privacy ensures that the global model learns from massive population data without ever moving raw patient data from local hospital servers. The Isolation Forest algorithm provides a specific technical solution for detecting multi-dimensional anomalies that simple threshold-based alerts miss.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 2 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 3 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 4 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 5 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 6 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 7 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 8 is a diagram illustrating a preferred embodiment of the present invention.

DETAILED DESCRIPTION

FIG. 1 illustrates the Federated GNN Digital Twin architecture within a HIPAA-compliant computation graph (100). FIG. 2 details the specific 12-domain graph structure, showing probabilistic edges between nodes like Mood (200) and Sleep. FIG. 3 maps the Federated Learning protocol, showing local models (301-304) sending encrypted, differentially private updates to a central server (300). FIG. 4 demonstrates the Isolation Forest anomaly detection separating normal states (400) from multi-dimensional anomalies (401). FIG. 5 details the GNN message-passing mechanism (510). FIG. 6 shows the edge computing module for local inference. FIG. 7 graphs the privacy-accuracy tradeoff benchmark. FIG. 8 visualizes the longitudinal anomaly detection timeline.

CLAIMS

What is claimed is:

1. A system for generating a mental health digital twin, comprising: a plurality of local edge computing servers deployed at distinct healthcare facilities; a central aggregation server; wherein each local edge computing server is configured to: instantiate a Graph Neural Network (GNN) comprising a plurality of nodes representing distinct psychiatric domains and a plurality of edges representing learned probabilistic dependencies between the domains; train the GNN on local patient data using a message-passing algorithm to update a hidden state vector for each node based on aggregated state vectors of neighboring nodes; apply a differential privacy noise injection algorithm to generate a privacy-preserving model gradient; and transmit the privacy-preserving model gradient to the central aggregation server; wherein the central aggregation server is configured to execute a secure aggregation protocol to update a global GNN model without accessing raw patient data; and wherein the local edge computing server executes an Isolation Forest algorithm on the hidden state vectors to detect a multi-domain psychiatric anomaly.

FIG. 1 - Federated GNN Digital Twin Architecture

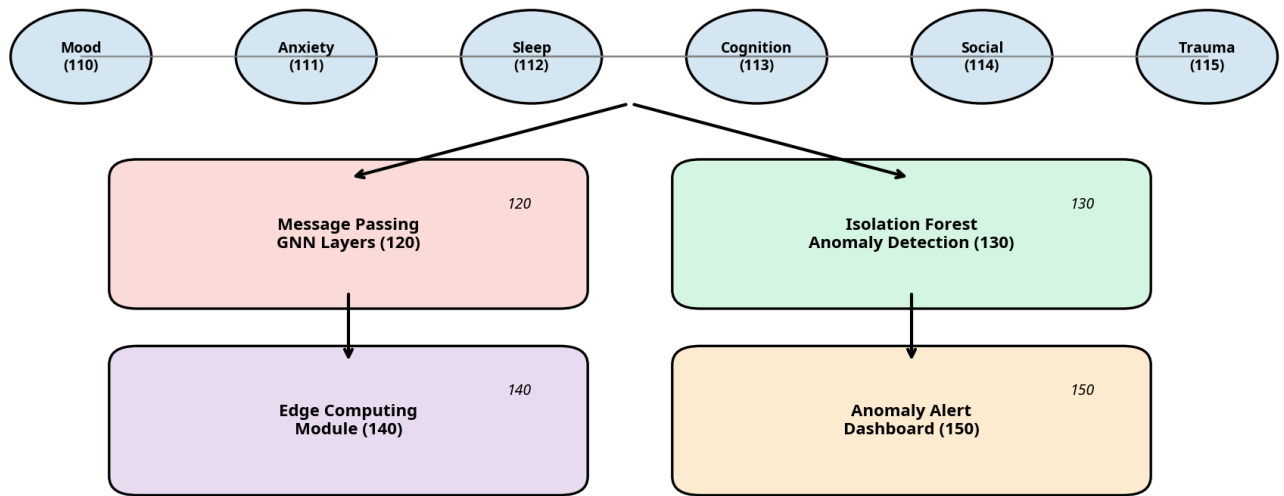
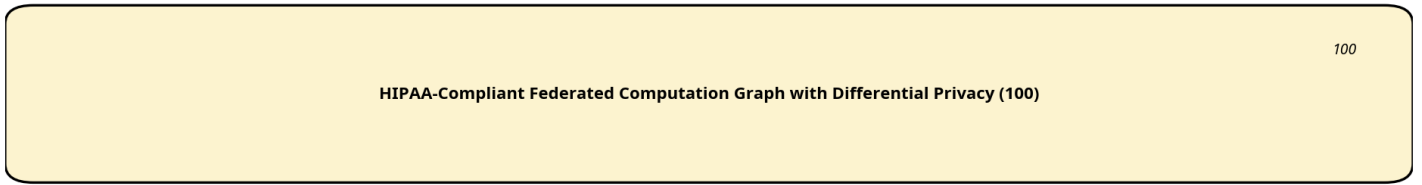


FIG. 2 - 12-Domain Mental Health Graph Structure

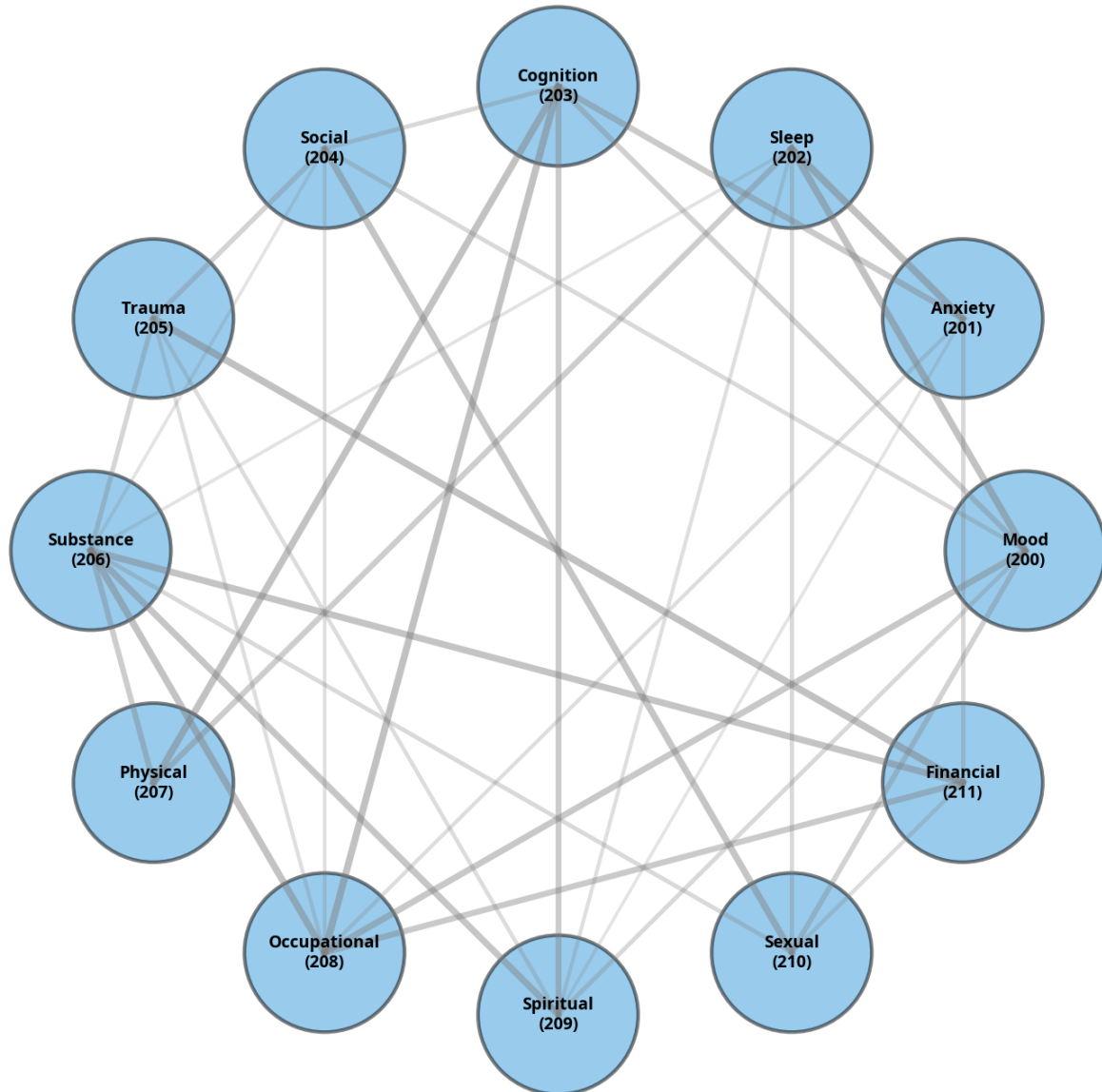


FIG. 3 - Federated Learning with Differential Privacy Protocol

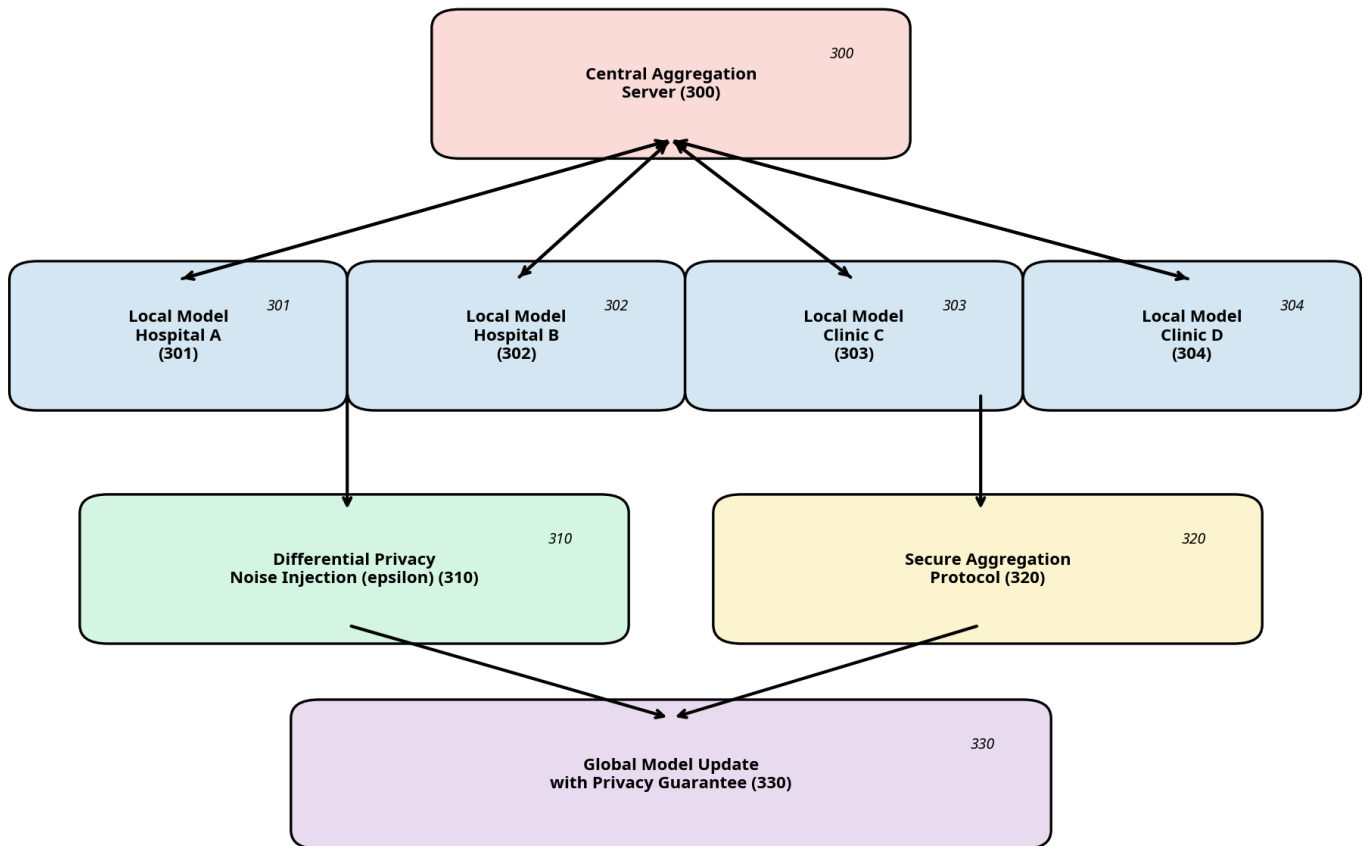


FIG. 4 - Isolation Forest Anomaly Detection in Digital Twin State

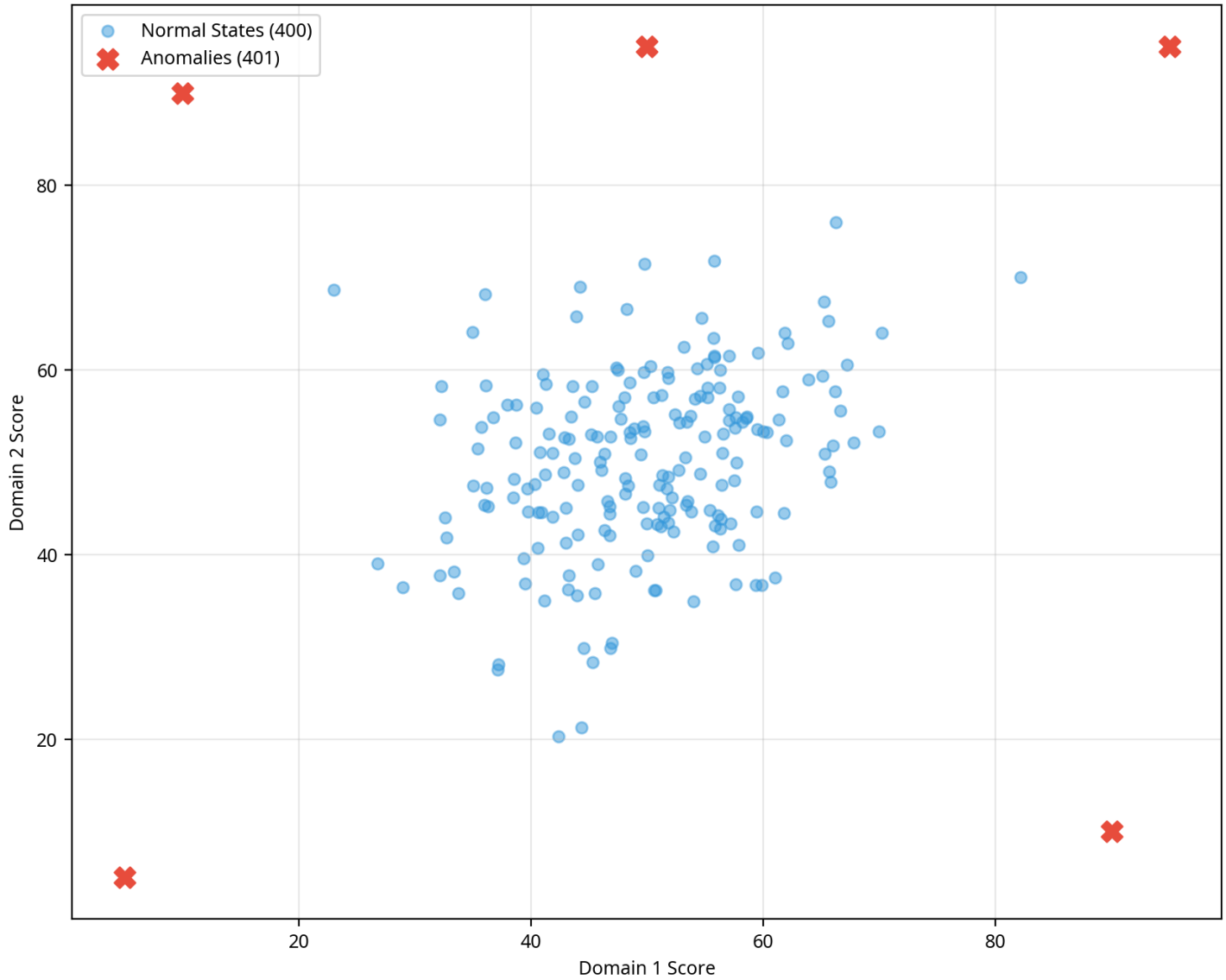


FIG. 5 - GNN Message Passing Mechanism

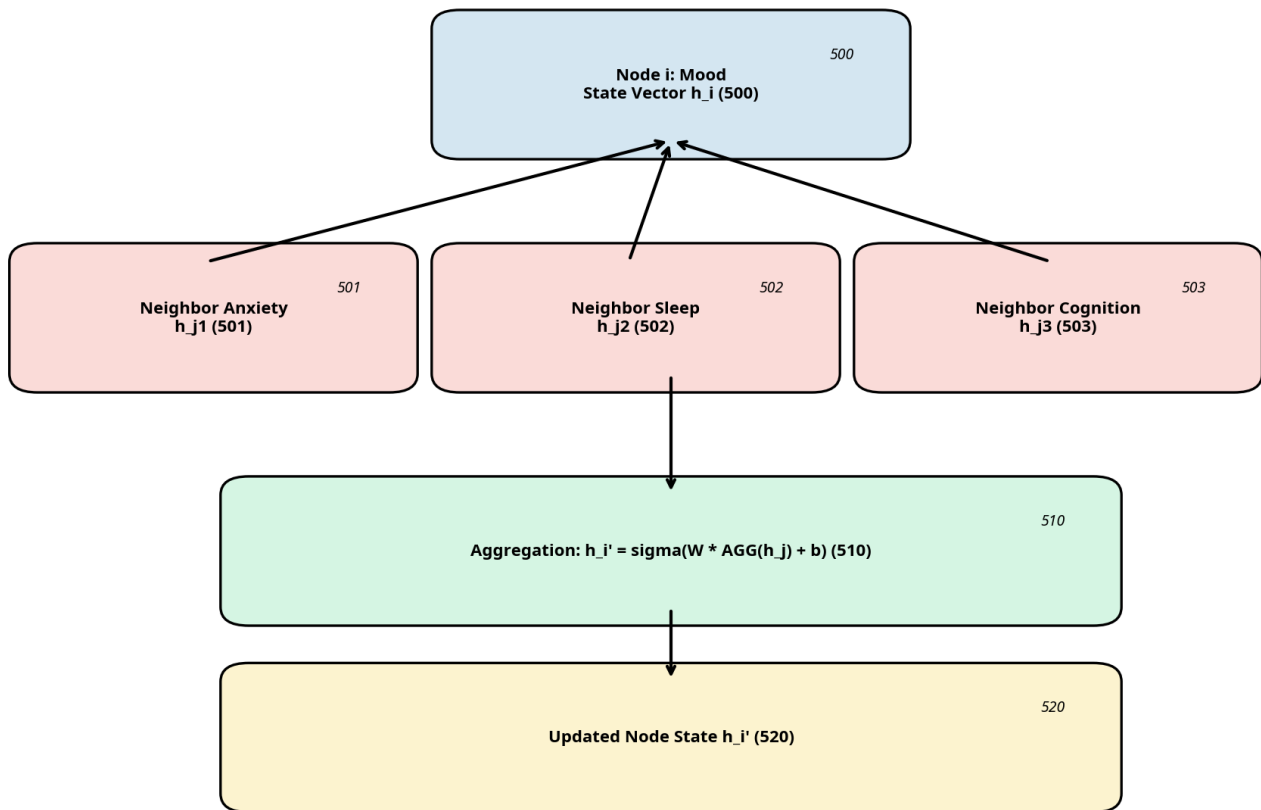


FIG. 6 - Edge Computing Module for Federated Updates

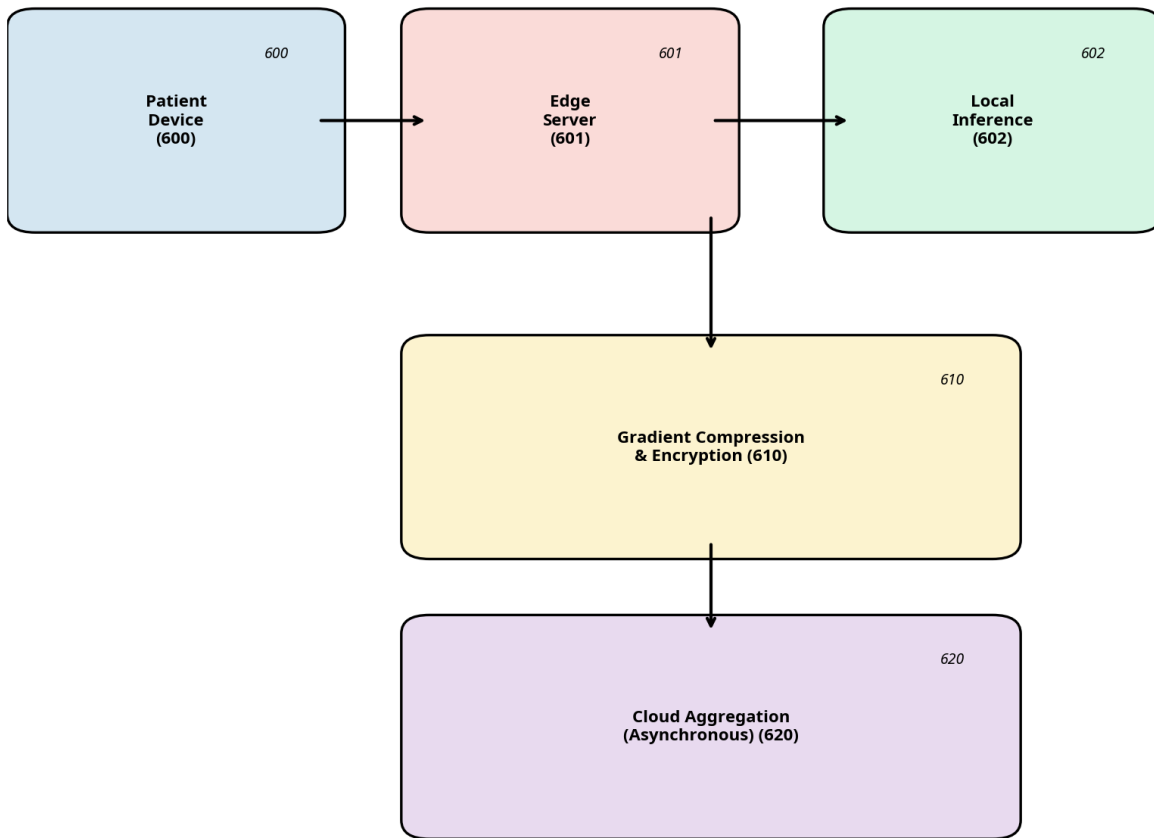
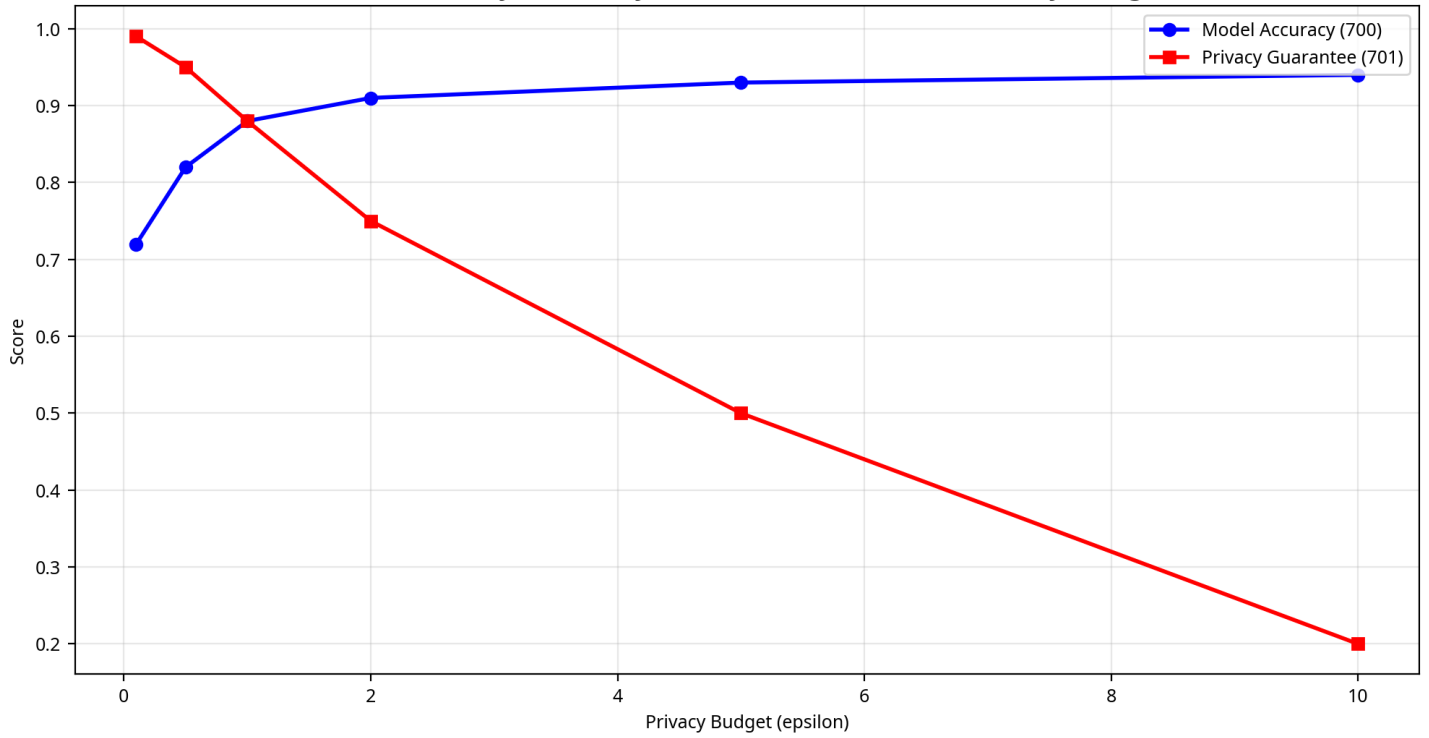


FIG. 7 - Privacy-Accuracy Tradeoff (Differential Privacy Budget)



UNITED STATES PATENT APPLICATION

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FIG. 8 - Longitudinal Anomaly Detection Timeline

